INCH-POUND

MIL-C-70572B (AR)
3 April 1991
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

CIRCUIT BOARD, BUILT IN TEST #2, UNIT 4: 9360952

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the AN/VPS-2A radar range computer Built-In-Test (BIT) board #2 assembly, referred to as the Unit 4 BB 2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-P-70565	-	Processor, Fire Control: 9360890

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1285

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STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-2000	-	Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (see 6.4)

9360952	-	BIT Board 2 Assembly, Computer, Range
9360954	-	BIT Board 2 Schematic, Computer, Range
9361339	-	Test Program Set Manual
9397373	-	Interface Device C
9397376-2	-	Cable Assembly, Polarized
9397501	-	Test Plug P1
9397502	-	Test Plug P2
9397503	-	Test Plug P15
9397504	-	Test Plug P3
9397562	-	Interface Device C W5 Printed Wiring Board Assembly
9397563	-	Cable Assembly IDCW4
10545465	-	Isolator

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

SPECIAL PACKAGING INSTRUCTION

PS9360952	-	Packaging of BIT Board 2 Assembly, Computer, Range

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection the specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS

ANSI/IPC-DW-425 - Design and End Product Requirements for Discrete Wiring Boards

(Application for copies should be addresed to ANSI, 1430 Broadway, New York, NY 10018).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9360952, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the Unit 4 BB 2 shall meet the requirements of MIL-F-13926 and ANSI/IPC/DW-425 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.2.1 ANSI/IPC-DW-425. IPC/DW-425/11 Type C1 plated through connection.

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3.3 Environmental service conditions.

3.3.1 Vibration. The Unit 4 BB 2 shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes.

Frequency Range	Amplitude
5 to 15 Hertz	0.06 inches double amplitude
15 to 25 Hertz	0.04 inches double amplitude
25 to 55 Hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 5-55-5 Hertz sweep time is 7 minutes 35 seconds ± 10%.

3.3.2 Shock. The Unit 4 BB 2 shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude (non-operating). The Unit 4 BB 2 shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The Unit 4 BB 2 shall meet the requirements of 3.4 and 3.5 when stabilized thermally at -40 degrees F and +155 degrees F.

3.3.4.2 Non-operating (storage). The Unit 4 BB 2 shall not suffer permanent damage when subjected to the constant temperature extremes of -60 degrees F for a period of 24 hours and +160 degrees F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The Unit 4 BB 2 shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.3.6 Electromagnetic interference. The class IC equipment requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram 9360954.

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3.5 Performance.

3.5.1 Functional description. The Unit 4 BB 2 shall monitor critical digital and analog signals within the Unit 4 AN/VPS-2A radar range computer. The digital signals shall be individually selected using digital multiplexers with address bytes received from the FCP CPU. Each digital signal is a pulse train. The pulses that make up a given signal shall be counted over a set time interval; this circuit is, therefore, a frequency counter. The count at the end of the time interval shall be output as an 8-bit byte to Unit 4 BB 1 for transmission to the FCP. At the FCP, the counts are compared to acceptable ranges in software. The analog signals shall be scaled and individually selected using an analog multiplexer with address bytes received from the FCP CPU. The scaled analog signals shall be converted to 8-bit digital bytes and output to Unit 4 BB 1 for transmission to the FCP. At the FCP, the digital values, representing analog voltage levels, are compared to acceptable ranges of values in software. The same signal that stops the digital count after the set time interval shall also start the analog to digital conversion, therefore, one "analog" byte shall be output with each digital byte to Unit 4 BB 1. The Unit 4 BB 2 also shall contain two digital multiplexers for selecting either hardware switches or software "switches" (bits) to activate a simulated dynamic target for radar computer test purposes. These multiplexer outputs shall go to two interface connectors, P1 and J1. Connector P1 shall distribute the switch outputs to other Unit 4 range computer boards and connector J1 shall send the outputs to Unit 4 BB 1 for transmission back to the FCP for end-around-test purposes.

3.5.2 Power requirements.

3.5.2.1 Input characteristics. Unit 4 BB 2 power shall be input at connector P1-17, 18 (+28 V) and P1-24, 25, 26 (+28 V RTN) with requirements as follows:

	Min	Typ	Max
Input voltage (volts)	16	24	32.0
Input current (amperes)	---	0.458	1.0
Ripple (mV p-p)	---	---	840.0

A 1N5647A diode shall suppress voltage spikes on the +28 V input line.

3.5.2.2 Output characteristics. With input per 3.5.2.1, the on-board dc-to-dc converter outputs shall be as follows with return reference to J1-14,15:

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Connector-pin	Voltage (volts)	Line regulation (mV)	Ripple (mVrms)
P1-21	+15.00±0.25	8 max	40 max
P1-22	-15.00±0.25	8 max	40 max
J1-4,5	+ 5.00±0.25	11 max	80 max

3.5.3 Clock requirements. The CLK 0 (500 kHz) input signal applied at J1-42 to J1-14,15 (rtn) shall have TTL voltage levels and a frequency of 500.000 kHz ±74 Hz.

3.5.4 Reset and timing.

3.5.4.1 Reset operation. With power and clock applied to the Unit 4 BB 2 per 3.5.2.1 and 3.5.3, respectively, a logic low (short to +5 V RTN) applied to J1-29 (RST) shall cause RST at P1-55 and CONV at P1-52 to go high and the following output signals to go low as long as the logic low is applied:

Signal	Connector and pins
C0 through C8	J1-31,49,21,46,18,32,7,27, and 26, respectively
CLK 4	P1-49
CLK 5	J2-2
CLK 8	P1-50
CLK 9	P1-51
QC	J2-3

Simultaneously, the following output signal lines shall tristate:

Signal lines	Connector and pins
AD1 through AD8	J1-45,47,19,12,8,6,33, and 30, respectively

3.5.4.2 Timing operation. With circuit power and clock applied per 3.5.2.1 and 3.5.3, respectively, the CLK 0 (500.000 kHz ±74 Hz) input applied at J1-42 shall divide by 16 to form CLK 4 (31.250 kHz ±5 Hz) at P1-49, divide by 32 to form CLK 5 (15.625 kHz ±3 Hz) at J2-2, divide by 256 to form CLK 8 (1.953 kHz ±1 Hz) at P1-50 and divide by 512 to form CLK 9 (976 ±1 Hz) at P1-51. When 10 CLK 9 pulses (5,120 of the 500 kHz pulses) have occurred since a reset per 3.5.4.1, CONV (P1-52) shall go low and the timing sequence shall halt until after another reset occurs. The timing sequence interval shall form the counter time base and be 10.24 ±0.01 milliseconds (5,120 pulses - 500 k ±74 pulses/second).

3.5.5 Digital multiplexer and counter.

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3.5.5.1 Initialization. With power, clock and RST (low) applied per 3.5.2.1, 3.5.3 and 3.5.4.1, respectively, holding RST per 3.5.4.1 (J1-29) low shall cause RST (P1-55) to go high and also keep the timing circuit of 3.5.4.2 from counting. $\overline{\text{CONV}}$ (P1-52) shall be high and the following signals shall be low:

a. C0 through C8 (J1-31,49,21,46,18,32,7,27, and 26, (respectively)
b. QC (J2-3)
c. CLK 4 (P1-49)
d. CLK 5 (J2-2)
e. CLK 8 (P1-50), and
f. CLK 9 (P1-51).

With RST brought high, the digital multiplexer and counter shall function per 3.5.5.2 and 3.5.5.3.

3.5.5.2 Digital multiplexer. Integrated circuits U4 and U5 together shall form a 16-line to 1-line digital multiplexer with common output at COUNTER MUX OUT (P1-48). The T3 - T10 (P1-45) signal (8 pulses every 50 microseconds) shall be divided by 8 by integrated circuit U8 to form signal QC before being applied to the multiplexer. The T_o TRIG (P1-46) signal shall be scaled down and checked by comparator U1 against a fixed reference voltage for sufficient amplitude before being applied to the multiplexer as signal COMP (P1-56). Individually applying each input signal of Table I (Digital multiplexer input characteristics) to connector P1 (unless generated internally) and applying its corresponding address code to RBR1 through RBR4 (J1-22,37,13,38), shall cause the selected multiplexer input signal to be output at COUNTER MUX OUT (P1-48). The input signal amplitudes, pulse widths and pulse repetition frequencies shall be per Table I. A high on RBR4 (J1-38) shall select multiplexer U5 and a low on RBR4 shall select multiplexer U4. The output of the multiplexer not selected shall be tristated.

3.5.5.3 Digital counter. The pulses on the COUNTER MUX OUT line (P1-48) shall be applied to a digital counter integrated circuit through a NAND gate controlled by the timing circuit of 3.5.4.2. The output of the gate shall be available at test point COUNT (P1-53). After a reset per 3.5.4.1 (see Figure 1, Digital Counter and A/D converter timing diagram), the counter shall count pulses on the COUNTER MUX OUT line for 10.24 $\pm$0.01 milliseconds, at which time $\overline{\text{CONV}}$ shall go low and cause the NAND gate output to go and remain high, inhibiting the count. The total count shall then be available at CO(lsb) through C8(msb) at J1-31,49,21,46,18,32,7,27 and 26, respectively. C8 shall act as a flag and go high if the count exceeds 255 and the counter recycles to zero. The binary count for each digital multiplexer input of Table I shall be per Table II (Digital counter output).

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TABLE I. Digital multiplexer input characteristics

Number	Input Digital Multiplexer Address RBR4 J1-38	RBR3 J1-13	RBR2 J1-37	RBR1 J1-22	Input Signal	Input Signal Source	Pulse Amplitude	Pulse Width at 50% Amplitude Points	Pulse Repetition Frequency (PRF)
1	0	0	0	0	CLK 4	Internal T.P. P1-49	TTL	Sq. WAVE	31,250±5 Hz
2	0	0	0	1	R	P1-43	TTL	0.8±0.1 us	3,996±1 Hz
3	0	0	0	1	R		TTL	133±10 ns	248±1 Hz
4	0	0	1	0	E/L TRIG OUT	P1-44	TTL	Variable 0.2 us to 0.6 us	19,979±1 Hz
5	0	0	1	1	R_E SCALED	P1-42	TTL	0.8±0.1 us	1,249±1 Hz
6	0	1	0	0	GND	Internal	---	---	---
7	0	1	0	1	GND	Internal	---	---	---
8	0	1	1	0	T3 T10 [1]	P1-45	TTL	2	[3]
9	0	1	1	0	QC (T3 T10-8)	U8-5	TTL	4.24±0.8 us	19,979±1 Hz
10	0	1	1	1	T_o TRIG	P1-46	16.6 Vmin. 25.1 Vmax.	0.35±0.20 us	19,979±1 Hz
11	0	1	1	1	COMP(T_o TRIG)	Internal	low: 0.4 Vmax	0.35±0.20 us	19,979±1 Hz

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TABLE I. Digital multiplexer input characteristics (continued)

Number	Input Digital Multiplexer Address RBR4 J1-38	RBR3 J1-13	RBR2 J1-37	RBR1 J1-22	Input Signal	Input Signal Source	Pulse Amplitude	Pulse Width at 50% Amplitude Points	Pulse Repetition Frequency (PRF)
12	1	0	0	0	PRF	P1-35	TTL	0.6±0.1 us	19,979±1 Hz
13	1	0	0	1	T2	P1-36	TTL	0.8±0.1 us	19,979±1 Hz
14	1	0	1	0	T11	P1-37	TTL	0.8±0.1 us	19,979±1 Hz
15	1	0	1	1	T1	P1-38	TTL	0.8±0.1 us	19,979±1 Hz
16	1	1	0	0	E/L TRIG	P1-39	TTL	Variable 0.4 us to 1.5 us	19,979±1 Hz
17	1	1	0	1	E/L COARSE	P1-40	TTL	Variable 35 us to 0.3 us	19,979±1 Hz
18	1	1	1	0	PREKNOCK (PRE-RANGE TRIGGER)	P1-41	TTL	Variable	19,979±1 Hz
19	1	1	1	1	CLK 5	Internal T.P. J2-2	TTL	SQ. WAVE	15,625±3 Hz

[1]Burst of 8 pulses in series.

[2]Pulse width of each pulse shall be 0.8±0.1 us.
Pulse spacing, leading edge to leading edge, shall be 1.06±0.1 us.

[3]Burst rate shall be 19,979±1 Hz.

T.P. means Test Point; where signals derived internally shall be measured.

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TABLE II. Digital counter output

Input Number	Input Signal	Decimal Count	Binary Count From C8	C7	C6	C5	C4	C3	C2	C1	C0	To C8	C7	C6	C5	C4	C3	C2	C1	C0
1	CLK 4	320±2	1	0	0	1	1	1	1	1	0	1	0	1	0	0	0	0	1	0
2	R	40±2	0	0	0	1	0	0	1	1	0	0	0	0	1	0	1	0	1	0
3	R	2±1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1
4	E/L TRIG OUT	205±2	1	1	0	0	1	0	1	1	0	0	1	1	0	0	1	1	1	1
5	R_E SCALED	13±2	0	0	0	0	0	1	0	1	1	0	0	0	0	0	1	1	1	1
6	GND	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
7	GND	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	T3 T10	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

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TABLE II. Digital counter output (continued)

Input Number	Input Signal	Decimal Count	Binary Count From C8	C7	C6	C5	C4	C3	C2	C1	C0	To C8	C7	C6	C5	C4	C3	C2	C1	C0
9	QC	205±2	0	1	1	0	0	1	0	1	1	0	1	1	0	0	1	1	1	1
10	T_o TRIG	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
11	COMP	205±2	0	1	1	0	0	1	0	1	1	0	1	1	0	0	1	1	1	1
12	PRF	205±2	0	1	1	0	0	1	0	1	1	0	1	1	0	0	1	1	1	1
13	T2	205±2	0	1	1	0	0	1	0	1	1	0	1	1	0	0	1	1	1	1
14	T11	205±2	0	1	1	0	0	1	0	1	1	0	1	1	0	0	1	1	1	1
15	T1	205±2	0	1	1	0	0	1	0	1	1	0	1	1	0	0	1	1	1	1
16	E/L TRIG	205±2	0	1	1	0	0	1	0	1	1	0	1	1	0	0	1	1	1	1
17	E/L COARSE	205±2	0	1	1	0	0	1	0	1	1	0	1	1	0	0	1	1	1	1
18	PREKNOCK	205±2	0	1	1	0	0	1	0	1	1	0	1	1	0	0	1	1	1	1
19	CLK 5	160±2	0	1	0	0	1	1	1	1	0	0	1	0	1	0	0	0	1	0

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3.5.6 Two quad 2-line to 1-line digital multiplexers. Integrated circuits U6 and U19 are quad 2-line to 1-line digital multiplexers. Each multiplexer is wired to select a 4-bit word from one of two sources and route it to its 4-bit output. With power applied per 3.5.2.1 and the S1 (J1-24) and SW2 (J1-48) select lines set as shown in the Conditions column of Table III (Quad 2-line to 1-line digital multiplexers U6 and U19), the signals shown in the Input column shall propagate within 29 nanoseconds to the outputs shown in the Output and Test Point columns of Table III.

3.5.7 Analog multiplexer and A/D converter. Signal timing shall be as shown in Figure 1.

3.5.7.1 A/D converter initialization. With power, clock and RST (low) applied per 3.5.2.1, 3.5.3, and 3.5.4.1, respectively, RST (P1-55) shall be high and the timing circuit of 3.5.4.2 shall not be counting. $\overline{\text{CONV}}$ (P1-52) shall be high and AD1 through AD8 (J1-45,47,19,12,8,6,33 and 30) shall be tristated.

3.5.7.2 Analog multiplexer and A/D converter operation. With the Unit 4 BB 2 initialized per 3.5.7.1 and the following inputs applied:

a. RBR4, 3, 2, and 1 (J1-38,13,37, and 22 respectively) set to 0000, and

b. All voltages V_{in} per Table IV (Analog multiplexer channels. Analog inputs and multiplexer outputs) connected to the Nominal Analog Inputs per Table IV (unless generated internally). The voltage levels at MUX OUT and A/D IN for multiplexer channel S1 shall be per Tables IV and V (Analog multiplexer channels. Analog inputs and multiplexer outputs; Analog multiplexer channels. A/D inputs and outputs), respectively. 10.24±0.01 milliseconds after $\overline{\text{RST}}$ (J1-29) is brought high, $\overline{\text{CONV}}$ shall go low to start the analog to digital conversion and within 40 microseconds maximum, channel S1 binary data per Table V shall be available at AD8 through AD1. Sequencing through the fifteen remaining multiplexer addresses with $\overline{\text{RST}}$ first low then high, shall yield the corresponding MUX OUT and A/D IN analog voltages and AD8 through AD1 binary data of Tables IV and V with timing per Figure 1.

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TABLE III. Quad 2-line to 1-line digital multiplexers U6 and U19

Input	Output	Conditions[1]
Open circuit[2]	SW1	SW2 low or SW2 high and S1 low
GND[2]	SW1	SW2 high and S1 either high or open
4S2-2	SW3	SW2 low or SW2 high and S1 high
S3	SW3	SW2 high and S1 low
4S2-5	SW4	SW2 low or SW2 high and S1 high
S4	SW4	SW2 high and S1 low
4S6-2	SW6	SW2 low
S6	SW6	SW2 high

Input	Test Point	Conditions[1]
+5 V RTN & ±15 V RTN	BS1	SW2 low
S1	BS1	SW2 high
S3	BS3	S1 low
4S2-2	BS4	S1 high
S4	BS4	S1 low
4S2-5	BS4	S1 high

[1]Low and high are 0.7 V max and 2.0 V min, respectively.

[2]Input is from the Unit 4 BB 2, not from an external source.

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TABLE IV. Analog multiplexer channels. Analog inputs and multiplexer outputs

Mux Channel	Mux Address A3 RBR4	A2 RBR3	A1 RBR2	A0 RBR1	Nominal Analog Input Connector - Pin	Analog Input V_{in} (in Vdc)	Mux Out (in Vdc)	
S1	0	0	0	0	+15 V/Internal	+14.750 to +15.250	+3.623 to +3.860	2.7< v <4.1
S2	0	0	0	1	+33 VDC/P1-9	+32.990 to +33.010	+4.602 to +4.766	1.4< v <2.0
S3	0	0	1	0	-33 VDC/P1-10	-32.990 to -33.010	-4.602 to -4.766	1.4< v <2.0
S4	0	0	1	1	+25 VDC/P1-11	+24.990 to +25.010	+3.486 to +3.611	1.4< v <2.0
S5	0	1	0	0	-25 VDC/P1-12	-24.990 to -25.010	-3.486 to -3.611	1.4< v <2.0
*S6	0	1	0	1	+15 VDC/P1-13	+14.990 to +15.010	+3.682 to +3.800	2.7< v <4.1
S7	0	1	1	0	-15 VDC/P1-14	-14.990 to 15.010	-3.682 to -3.800	2.7< v <4.1
S8	0	1	1	1	+6.5 VDC/P1-15	+6.490 to +6.510	+3.212 to +3.288	4.7< v <6.3
S9	1	0	0	0	+5 VDC/P1-31	+4.990 to +5.010	+2.470 to +2.530	4.7< v <6.3
S10	1	0	0	1	K1 RELAY +28 V/P1-16	+27.990 to +28.010	+3.904 to +4.044	1.4< v <2.0
S11	1	0	1	0	+28 V/P1-17,18	+27.990 to +28.010	+3.904 to +4.044	2.7< v <4.1
*S12	1	0	1	1	SUM SIGNAL/P1-19	+3.490 to +3.510	+1.727 to +1.773	4.7< v <6.3
S13	1	1	0	0	GND/Internal	-0.010 to +0.010	-0.010 to +0.010	-.5< v <.5
S14	1	1	0	1	+5/Internal	+4.750 to +5.250	+4.750 to +5.250	4.7< v <5.3
S15	1	1	1	0	GND/Internal	-.0.010 to +0.010	-0.010 to +0.010	-.5< v <.5
S16	1	1	1	1	-15 V/Internal	-14.750 to -15.250	-3.623 to -3.860	-4.7< v <3.1

*12 v in

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TABLE V. Analog multiplexer channels. A/D inputs and outputs

Mux Channel	Mux Address A3 RBR4	A2 RBR3	A1 RBR2	A0 RBR1	A/D IN (in Vdc)	A/D Output Word From AD8 MSB	AD7	AD6	AD5	AD4	AD3	AD2	AD1 LSB	To AD8 MSB	AD7	AD6	AD5	AD4	AD3	AD2	AD1 LSB
S1	0	0	0	0	+3.613 to +3.860	1	1	0	1	1	0	1	1	1	1	1	0	0	1	0	0
S2	0	0	0	1	+4.592 to +4.766	1	1	1	1	0	1	0	0	1	1	1	1	1	1	0	0
S3	0	0	1	0	-4.592 to -4.766	0	0	0	0	0	1	1	0	0	0	0	0	0	1	0	0
S4	0	0	1	1	+3.476 to +3.611	1	1	0	1	1	0	0	0	1	1	0	1	1	1	1	0
S5	0	1	0	0	-3.476 to -3.611	0	0	1	0	1	0	0	0	0	0	1	0	0	0	1	0
S6	0	1	0	1	+3.672 to +3.800	1	1	0	1	1	1	0	1	1	1	0	1	0	0	1	1
S7	0	1	1	0	-3.672 to -3.800	0	0	1	0	0	0	1	1	0	0	0	1	1	1	0	1
S8	0	1	1	1	+3.202 to +3.288	1	1	0	1	0	0	0	1	1	1	0	1	0	1	0	1
S9	1	0	0	0	+2.460 to +2.530	1	0	1	1	1	1	1	0	1	1	0	0	0	0	1	0
S10	1	0	0	1	+3.894 to +4.044	1	1	1	0	0	0	1	0	1	1	1	0	1	0	0	1
S11	1	0	1	0	+3.894 to +4.044	1	1	1	0	0	0	1	0	1	1	1	0	1	0	0	1
S12	1	0	1	1	+1.717 to +1.773	1	0	1	0	1	0	1	1	1	0	1	0	1	1	1	0
S13	1	1	0	0	-0.020 to +0.010	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0
S14	1	1	0	1	+4.740 to +5.250	1	1	1	1	1	0	0	0	1	1	1	1	1	1	1	1
S15	1	1	1	0	-0.020 to +0.010	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0
S16	1	1	1	1	-3.613 to -3.860	0	0	1	0	0	1	0	1	0	0	0	1	1	1	0	0

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3.5.8 Input/output requirements. The Unit 4 BB 2 inputs shall be as listed in Tables VI (P1 inputs) and VII (J1 inputs). The Unit 4 BB 2 outputs shall be as listed in Tables VIII (P1 outputs) and IX (J1 outputs). Input/output pin designations shall be as listed in Tables VI, VII, VIII, and IX. These tables contain low logic input voltages (LLIVs), high logic input voltages (HLIVs), low logic input currents (LLIIs) and high logic input currents (HLIIs).

3.5.9 Test points. The Unit 4 BB 2 shall contain test points as listed in Table X (Test points).

3.5.10 Spare connector pins. The Unit 4 BB 2 shall contain spare pins on connectors P1, J1, and J2 as listed in Table XI (Spare pins).

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

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TABLE VI. P1 inputs

CONNECTOR/ PIN	SIGNAL	DESCRIPTIVE NOTES	MAX LLIV	MIN HLIV	MAX LLII	MAX HLII
P1-3	4S2-2		0.7 V	2 V	-0.8 ma	40 ua
P1-4	4S2-5		0.7 V	2 V	-0.8 ma	40 ua
P1-5	4S6-2		0.7 V	2 V	-0.4 ma	20 ua
P1-9	+33 VDC					
P1-10	-33 VDC					
P1-11	+25 VDC					
P1-12	-25 VDC					
P1-13	+15 VDC					
P1-14	-15 VDC					
P1-15	+6.5 VDC					
P1-16	+28 V K1 RELAY					
P1-17	+28 VDC	+28 V PWR INPUT				
P1-18	+28 VDC	+28 V PWR INPUT				
P1-19	SUM SIG					
P1-31	+5 VDC					
P1-24	+28 RTN	+28 V PWR RTN				
P1-25	+28 RTN	+28 V PWR RTN				
P1-26	+28 RTN	+28 V PWR RTN				
P1-35	PRF		0.7 V	2 V	-0.4 ma	20 ua
P1-36	T2		0.7 V	2 V	-0.4 ma	20 ua
P1-37	T11		0.7 V	2 V	-0.4 ma	20 ua
P1-38	T1		0.7 V	2 V	-0.4 ma	20 ua
P1-39	E/L TRIG		0.7 V	2 V	-0.4 ma	20 ua
P1-40	E/L COARSE		0.7 V	2 V	-0.4 ma	20 ua
P1-41	PREKNOCK		0.7 V	2 V	-0.4 ma	20 ua
P1-42	RE SCALED		0.7 V	2 V	-0.4 ma	20 ua
P1-43	R		0.7 V	2 V	-0.4 ma	20 ua
P1-44	E/L TRIG OUT		0.7 V	2 V	-0.4 ma	20 ua
P1-45	T3-T10		0.7 V	2 V	-0.4 ma	20 ua
P1-46	TO TRIG		0.7 V	2 V	-1.6 ma	40 ua

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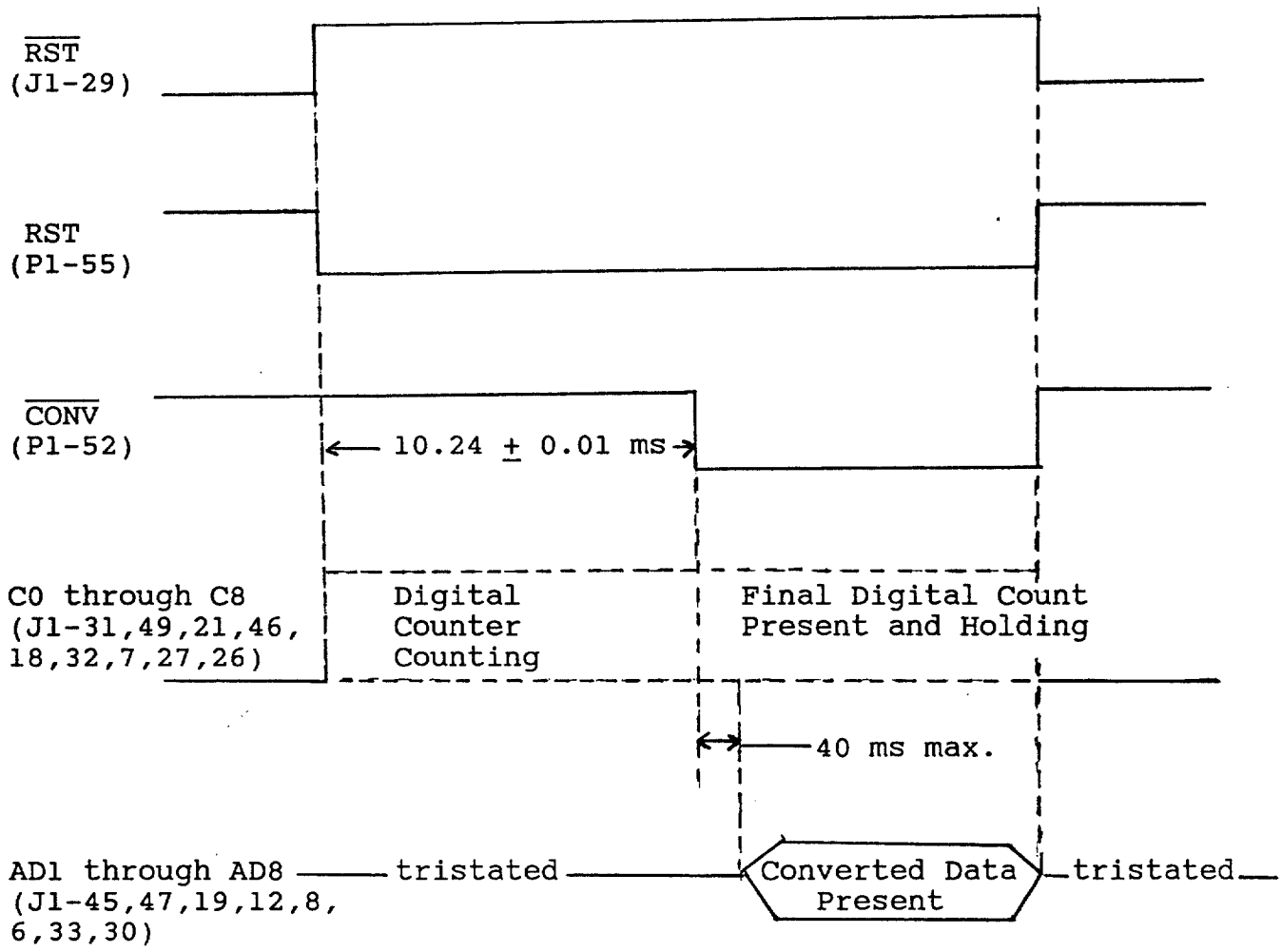


FIGURE 1. Digital counter and A/D converter timing diagram

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TABLE VII. J1 inputs

CONNECTOR/ PIN	SIGNAL	MAX LLIV	MIN HLIV	MAX LLIV	MAX HLII
J1-13	RBR3	0.7 V	2.4 V	-0.8 ma	50 ua
J1-22	RBR1	0.7 V	2.4 V	-0.8 ma	50 ua
J1-23	S3	0.7 V	2.0 V	-0.8 ma	40 ua
J1-24	S1	0.7 V	2.0 V	-1.2 ma	60 ua
J1-25	S4	0.7 V	2.0 V	-0.8 ma	40 ua
J1-29	RST	0.7 V	2.0 V	-2.4 ma	160 ua
J1-37	RBR2	0.7 V	2.4 V	-0.8 ma	50 ua
J1-38	RBR4	0.7 V	2.4 V	-0.8 ma	50 ua
J1-42	CLK 0 (500 kHz)	0.7 V	2.0 V	-1.6 ma	40 ua
J1-48	SW2	0.7 V	2.0 V	-0.8 ma	40 ua
J1-50	S6	0.7 V	2.0 V	-0.4 ma	20 ua

TABLE VIII. P1 outputs

CONNECTOR/ PIN	SIGNAL	DESCRIPTIVE	MAX LLOV	MIN HLOV	MAX LLOI	MAX HLOI
P1-1	GND					
P1-2	SW1	#	0.4 V	2.4 V	4 ma	-400 ua
P1-6	SW3	#	0.4 V	2.5 V	4 ma	-400 ua
P1-7	SW4	#	0.4 V	2.5 V	4 ma	-400 ua
P1-8	SW6	#	0.4 V	2.5 V	4 ma	-400 ua

Signal also outputted on J1

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TABLE IX. J1 outputs

CONNECTOR/ PIN	SIGNAL	DESCRIPTIVE NOTES	MAX LLOV	MIN HLOV	MAX LLOI	MAX HLOI
J1-3	R	STRAIGHT THROUGH CONNECTION				
J1-4	BIT+5 V					
J1-5	BIT+5 V					
J1-6	AD6		.4 V	2.4 V	3.2 ma	-.5 ma
J1-7	C6		.4 V	2.5 V	4.0 ma	-400 ua
J1-8	AD5		.4 V	2.4 V	3.2 ma	-.5 ma
J1-10	SW6	#	.4 V	2.5 V	4.0 ma	-400 ua
J1-12	AD4		.4 V	2.4 V	3.2 ma	-.5 ma
J1-14	BIT+5 V RET					
J1-15	BIT+5 V RET					
J1-18	C4		.4 V	2.5 V	4.0 ma	-400 ua
J1-19	AD3		.4 V	2.4 V	3.2 ma	-.5 ma
J1-20	SW3	#	.4 V	2.5 V	4.0 ma	-400 ua
J1-21	C2		.4 V	2.5 V	4.0 ma	-400 ua
J1-26	C8		.4 V	2.5 V	4.0 ma	-400 ua
J1-27	C7		.4 V	2.4 V	3.2 ma	-320 ua
J1-30	AD8		.4 V	2.5 V	3.2 ma	-.5 ma
J1-31	C0		.4 V	2.5 V	4.0 ma	-400 ua
J1-32	C5		.4 V	2.5 V	4.0 ma	-400 ua
J1-33	AD		.4 V	2.5 V	3.2 ma	-.5 ma
J1-40	SW1	#	.4 V	2.4 V	4.0 ma	-400 ua
J1-43	SW4	#	.4 V	2.5 V	4.0 ma	-400 ua
J1-45	AD1		.4 V	2.4 V	3.2 ma	-.5 ma
J1-46	C3		.4 V	2.5 V	4.0 ma	-400 ua
J1-47	AD2		.4 V	2.4 V	3.2 ma	-.5 ma
J1-49	C1		.4 V	2.5 V	4.0 ma	-400 ua

Signal also outputted on P1

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TABLE X. Test points

CONNECTOR/ PIN	SIGNAL	DESCRIPTIVE NOTES	NORMAL RANGE	MAX LLOV	MIN HLOV	MAX LLOI	MAX HLOI
P1-21	+15 V BIT		+15 v				
P1-22	-15 V BIT		-15 v				
P1-23	MUX OUT	ANALOG OUTPUT	-5 v, +5 v				
P1-33	A/D IN	ANALOG OUTPUT	-5 v, +5 v				
P1-48	COUNTER MUX OUT		0-5 v	0.4 v	2.4 v	4 ma	-1 ma
P1-49	CLK4		0-5 v	0.4 v	2.5 v	3.6 MA	-380 ua
P1-50	CLK8		0-5 v	0.4 v	2.5 v	3.6 ma	-380 ua
P1-51	CLK9		0-5 v	0.4 v	2.5 v	4.0 ma	-400 ua
P1-52	CONV		0-5 v	0.4 v	2.5 v	3.1 ma	-320 ua
P1-53	COUNT		0-5 v	0.4 v	2.5 v	2.4 ma	-360 ua
P1-54	RBR4		0-5 v	0.4 v	2.5 v	3.6 ma	-380 ua
P1-55	RST		0-5 v	0.4 v	2.5 v	1.6 ma	-280 ua
P1-56	COMP		0-5 v	0.4 v	2.4 v	6.0 ma	-480 ua
J2-2	CLK5		0-5 v	0.4 v	2.5 v	3.6 ma	-380 ua
J2-3	QC		0-5 v	0.4 v	2.5 v	4.0 ma	-400 ua
J2-4	BS1		0-5 v	0.4 v	2.5 v	3.6 ma	-380 ua
J2-5	BS3		0-5 v	0.4 v	2.5 v	3.6 ma	-380 ua
J2-6	BS4		0-5 v	0.4 v	2.5 v	3.6 ma	-380 ua

TABLE XI. Spare pins

CONNECTOR-PIN		
P1-27-30	J1-1	J2-1
P1-32	J1-2	J2-7-10
P1-20	J1-9	---
P1-34	J1-11	---
P1-47	J1-16	---
---	J1-17	---
---	J1-28	---
---	J1-34-36	---
---	J1-41	---
---	J1-44	---
---	J1-39	---
---	---	---

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3.7 Workmanship. The workmanship requirements of MIL-F-13926 shall apply.

3.8 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement task F of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling (see 4.5) (if required).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

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4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) Unit 4 BB 2 for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table XII (Functional Test) and Table XIII (Special Sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogenous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

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4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table XII shall be performed on a 100% basis.

TABLE XII. Functional tests
(100% Inspection)

Characteristic	Requirement	Test Procedure
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

4.5 Special sampling. One Unit 4 BB 2 out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table XII and shall then meet the tests in Table XIII. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE XIII. Special sampling

Characteristic	Requirements	Test Procedure
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

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4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2 the test equipment listed in Table XIV (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

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TABLE XIV. Test equipment

Item Number		Description
1.	Interface Device C	9397373 with PWB W5 (9397562) W4 Cable (9397563), and test cable W6 (9397376-2)
2.	Digital Card Tester	Gen Rad 2225A (AN/USM-465A)
3.	Com Code Program Tape	Version 2.0
4.	Test Program Set Manual	TPS A9361339
5.	Test Program Tape	TPT 9391339
6.	Oscilloscope	Input impedance of 1 Mohm or greater, single sweep capability, with probe

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The Unit 4 BB 2 assembly shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the Unit 4 BB 2 shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 Hertz.

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After the application of all shock impulses, the Unit 4 BB 2 shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance.

4.7.3.1 Reset and timing. The reset and timing shall be tested to meet the requirements of 3.5.4.

4.7.3.2 Digital multiplexer and counter. The digital multiplexer and counter shall be tested to meet the requirements of 3.5.5.

4.7.3.3 Two quad 2-line to 1-line digital multiplexers. The two quad 2-line to 1-line digital multiplexers shall be tested to meet the requirements of 3.5.6.

4.7.3.4 Analog multiplexer and A/D converter. The analog multiplexer and A/D converter shall be tested to meet the requirements of 3.5.7.

4.7.3.5 Input/output. The input/output shall be tested to meet the requirements of 3.5.8.

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4.7.3.6 Test points. The test points shall be tested to meet the requirements of 3.5.9.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.7

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet PS9360952.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. "The Unit 4 BB 2 provides the ability to monitor critical digital and analog signals within the Unit 4 range computer, while the system is in an operational mode or test mode, and output these signals to Unit 4 B1 for transmission to the Fire Control Processor (FCP). Refer to Figure 2 (Unit 4 BB 2 functional block diagram)."

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Applicable stock number.
c. First article sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certificate of conformance for each lot or shipment of product.
g. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423, in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now cognizance of ARDEC.

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6.5 Subject term (key word) listing.

Built-in-Test (BIT)
Fire Control System
Fire Control Processor
Product Improved Vulcan Air Defense System (PIVADS)
Radar Set AN/VPS-2A

6.6 Changes from previous issue. The previous issues were not forwarded to Naval Publications and Forms Center (NPFC) for publication. This issue is the initial public release of unlimited distribution.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A771)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-70572B (AR)	3 APRIL 1991

3. DOCUMENT TITLE

CIRCUIT BOARD, BUILT IN TEST # 2, UNIT 4: 9360952

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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